

ABHISHEK SEN

Robotics & Autonomous Systems Researcher | B.Tech in Electronics & Communication Engineering

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RESEARCH INTERESTS

Autonomous Robotics & Safe Control: Control Barrier Functions (CBFs), Gaussian Process (GP) regression, Provable Collision Avoidance, Quadratic Programming (QP), Motion Planning under Epistemic Uncertainty, Model Predictive Control (MPC).

Perception & State Estimation: 3D/4D LiDAR SLAM, Rao-Blackwellized Particle Filtering, Pseudo RGB-D Vision (Foundation Depth Models), Sensor Fusion (Extended Kalman Filters, Factor Graphs), Visual Servoing, Autonomous Aerial Vehicles (UAVs) & Swarms.

EDUCATION

Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Kancheepuram Chennai, India

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering

Nov 2022 – May 2026

- **Cumulative GPA: 8.50 / 10.0** (First Class with Distinction)
- **Undergraduate Capstone Thesis:** *Probabilistic Safe Navigation in Obstacle-Cluttered Dynamic Environments: A Gaussian Process-Enabled Control Barrier Function Approach* (Co-advised with IIT Madras)
- **Relevant Coursework:** Computer Vision, Data Structures and Algorithms, Object-Oriented Programming, Control Systems, Signals and Systems, Probability and Stochastic Processes, Embedded Systems, Microprocessors & Microcontrollers.

Govt. Model Higher Secondary School, Sehore

Madhya Pradesh, India

Higher Secondary Certificate (Class XII) — Physics, Chemistry, Mathematics

2020 – 2021

- **Grade: A+ | State Merit Scholar:** Ranked in **Top 1% statewide** across Madhya Pradesh Board (MPBSE).

Govt. Higher Secondary School, Ramnagar, Sehore

Madhya Pradesh, India

Secondary School Certificate (Class X)

2018 – 2019

- **Grade: A+ | District Rank 1:** Conferred top academic distinction by the **District Collector** and Member of Legislative Assembly (MLA).

UNDERGRADUATE CAPSTONE THESIS

Probabilistic Safe Navigation in Obstacle-Cluttered Dynamic Environments: A GP-CBF-QP Approach

Author: Abhishek Sen | Advisers: Dr. Selvajyothi K (IIITDM) & Dr. Satadal Ghosh (IIT Madras)

May 2026

- **Mathematical Formulation & Proofs:** Formulated a unified **GP-CBF-QP** framework bridging Bayesian non-parametric machine learning with non-linear control theory. Derived an analytical concentration inequality over squared exponential GP kernels to infer unknown velocity vector fields of dynamic obstacles online with calibrated epistemic uncertainty bounds. Formulated a formal mathematical proof establishing forward invariance and probabilistic safety: $\mathbb{P}[h(x(t), x_o(t)) \geq 0, \forall t \geq 0] \geq 1 - \delta$.
- **Simulation Benchmarking:** Validated across 200 rigorous simulation trials, demonstrating an **81% reduction in collision rate** compared to reactive baseline planners and **72% reduction** over deterministic CBFs, while retaining trajectory efficiency within 8% of the unconstrained optimum.
- **Real-Time Nano-UAV Hardware Deployment:** Implemented and physically deployed the entire closed-loop control pipeline onto **Crazyflie 2.1 nano-quadrators** in an obstacle-dense arena with dynamic drone obstacles. Integrated Ultra-Wideband (UWB) Loco Positioning and real-time IMU sensor fusion at 20 Hz, achieving a **100% mission task completion rate** across 34 physical flight experiments with an average Quadratic Program (QP) solve time of 1.20 ± 0.15 ms.

RESEARCH & PROFESSIONAL EXPERIENCE

Robotics Engineer

Jun 2026 – Present

RobotSpace

Hyderabad, India

- Spearheading autonomous navigation and path planning for an industrial Autonomous Outdoor UGV carrying a **50 kg payload** with **2 cm positional accuracy** via RTK-GPS correction and IMU-wheel odometry fusion.
- Developed the core software stack utilizing **ROS 2, PX4 Autopilot, MAVSDK**, and **Gazebo Harmonic**; engineered a custom Ground Control Station (GCS) telemetry interface with mobile/desktop control via REST APIs, Docker, and SQLite.
- Formulated an adaptive nighttime navigation pipeline combining dynamic obstacle avoidance with ambient lux intensity estimation.

Project Assistant (Autonomous Swarm Drones with Dynamic Motion Planning) *Oct 2025 – May 2026*
Guidance, Navigation & Control (GNC) Lab, IIT Madras [\[Research Paper Link\]](#) | Chennai, India

- Developed decentralized dynamic collision avoidance for autonomous drone swarms navigating in cluttered GPS-denied environments utilizing an Ultra-Wideband (UWB) Loco-positioning setup.
- Embedded the full navigation and trajectory evaluation stack onto an **NVIDIA Jetson Orin Nano** companion computer paired with a **Livox 4D LiDAR** for dense real-time point-cloud acquisition.
- Benchmarked the Generalized Shape Expansion (GSE) algorithm against traditional sampling-based baselines (RRT*), exhibiting superior computational efficiency and obstacle clearance in dynamic multi-agent regimes.
- Designed multi-sensor fusion pipelines using Extended Kalman Filters (EKF) and Particle Filters to blend 4D LiDAR odometry and high-frequency IMU telemetry for robust state estimation under aggressive maneuvers.

Research Intern (Autonomous Systems Laboratory) *May 2024 – Jul 2024*
Autonomous Systems Laboratory, IIITDM Kancheepuram (Under Dr. Ram Prasad Padhy, IIT Bhubaneswar)
Chennai, India

- Programmed a real-time ROS 2 driver and perception interface for a **Velodyne VLP-16 3D LiDAR** mounted on an autonomous ground mobile robot.
- Implemented Rao-Blackwellized Particle Filtering (GMapping SLAM), incorporating adaptive sampling distributions and selective resampling thresholds to mitigate particle depletion during large-scale indoor mapping.
- Converted raw sensor streams from the benchmark **KITTI Vision Benchmark Suite** into structured ROS 2 bag formats, optimizing transform pipelines (TF2) and evaluating drift metrics against ground truth.
- Engineered and hardware-tested a stereo-vision spatial object tracker to generate reactive bounding volumes for local obstacle avoidance.

AI Research and Development Intern *May 2025 – Oct 2025*
32Mins Digital Consultancy Services, IIT Madras Research Park Chennai, India

- Developed edge-based computer vision models on embedded Jetson platforms to track human eye gaze, facial micro-gestures, and posture metrics for real-time focus estimation.
- Deployed low-latency speech recognition and voice feedback pipeline to enable natural voice interaction directly on embedded microprocessors.

KEY ROBOTICS PROJECTS (SOURCE CODE & DEMOS ON GITHUB)

Pseudo RGB-D SLAM ROS 2 Package (Foundation Metric Depth Integration) [\[GitHub Repository\]](#)

- Designed a modular ROS 2 architecture replacing physical depth sensors with zero-shot monocular depth estimation foundation models (**Depth Anything V2** and **UniDepth**).
- Streamed monocular RGB frames from a CSI Sony IMX219 sensor on an **NVIDIA Jetson Orin Nano Super**, inferring aligned metric depth maps in real time and streaming pseudo-RGB-D point clouds into **ORB-SLAM2**.
- Generated dense 3D visual odometry trajectories and feature point-cloud maps rendered in RViz2; benchmarked on TUM RGB-D datasets.

Robust Navigation & Trajectory Control for Autonomous Mobile Robots [\[GitHub Repository\]](#)

- Formulated a complete custom navigation pipeline for differential-drive mobile robots (TurtleBot3) bypassing traditional Nav2 costmaps.
- Implemented B-Spline interpolation for global waypoint smoothing to enforce velocity and curvature continuity (C^2 smoothness).
- Integrated a look-ahead Pure Pursuit tracking controller with a fast sequential index search; coupled with an Artificial Potential Field (APF) reactive layer utilizing 2D planar LiDAR scans to dynamically deviate around unexpected obstacles.

Autonomous Docking & Visual Servoing AMR System [\[GitHub Repository\]](#)

- Developed an autonomous docking ROS 2 package executing vision-guided closed-loop alignment towards specialized docking targets in Gazebo simulation and hardware.
- Fused monocular bearing angles with planar LiDAR radial depth to compute 6-DOF transform offsets in TF2; regulated drive wheels via proportional-integral servo controllers to achieve sub-centimeter terminal docking accuracy while managing obstacle clearance.

ANAV: Autonomous GPS-Denied Drone for Martian Exploration (ISRO Robotics Challenge) *URSC Bengaluru*

- Designed an autonomous quadrotor UAV customized with an Intel RealSense D435i depth sensor, optical flow, and an onboard companion computer for GPS-denied planetary exploration.
- Implemented visual SLAM and YOLOv8 object/boundary detection integrated via ROS 2 and MAVROS to publish real-time local position targets to the flight controller, executing autonomous exploration and precision landing.

3D Dynamic Object Tracking with LiDAR & ArduPilot Simulation [\[GitHub Repository\]](#)

- Implemented real-time Euclidean clustering and 3D bounding box tracking on dense point clouds to predict dynamic obstacle vectors.
- Integrated ArduPilot SITL with Gazebo Classic for software-in-the-loop autonomous mission validation and reactive waypoint diversion.

HONORS, AWARDS & DISTINCTIONS

- **Top National Finalist, ISRO Robotics Challenge 2025:** Selected among top national teams; demonstrated the autonomous GPS-denied UAV “ANAV” at URSC Bengaluru and at the **National Space Day Ceremony at Bharat Mandapam, New Delhi**. Technology was reviewed and directly commended by **Dr. S. Somanath** (former ISRO Chairman) and **Dr. V. Narayanan** (ISRO Chairman); interacted directly with the four **Gaganyaan Astronaut-Designates** (Group Captain Angad Pratap et al.).
- **All India Rank (AIR) 8, SAE Aerothon 2025:** Secured 8th rank nationwide for Technical Design Report and flight presentation as Captain of Aerobotics Team Astra.
- **Winner, Best Challenger Award & AIR 21, SAE Aerothon 2023:** Placed 1st in the Challenger Category and achieved AIR 21 in report documentation.
- **Autonomous Mission Completion, IIT Roorkee:** One of only 4 teams nationwide to successfully complete both manual and autonomous obstacle-course flight regimes.
- **National Cadet Corps (NCC) ‘C’ Certificate with ‘A’ Grade:** Highest qualification awarded in NCC, demonstrating disciplined military leadership, endurance, and civic service.
- **State Merit Scholar (Class XII, 2021):** Conferred merit scholarship by the Secretary of MPBSE for ranking in the top 1% of students statewide.
- **District Rank 1 (Class X, 2019):** Felicitated with special merit awards by the **District Collector** and Member of Legislative Assembly (MLA) for highest scholastic achievement.
- **Cultural & Academic Awards:** Recipient of the *Pratibha Protsahan Award* (M.P. Rashtrabhasha Prachar Samiti), State Tourism Board Award, and State Biodiversity Conservation Award.

LEADERSHIP & ACADEMIC RESPONSIBILITIES

Team Captain & ECE Lead, Aerobotics Team Astra, IIITDM Kancheepuram

Jul 2023 – Feb 2026

- Led a 10-member multidisciplinary engineering team across four successive national rounds of the ISRO Robotics Challenge.
- Directed flight controller configuration, propulsion calculations, power distribution PCB designs, and companion computers (Jetson / RPi).

Senior Cadet, National Cadet Corps (NCC), Ministry of Defence, India

Nov 2022 – Feb 2025

- Completed rigorous annual training camps, rifle firing drills, and national social service camps; awarded Grade ‘A’ in ‘C’ Certificate examination.

TECHNICAL SKILLS

Control & Robotics:	ROS 2 (Humble/Iron), ROS 1 (Noetic), PX4 Autopilot, ArduPilot, MAVROS, MAVSDK, Control Barrier Functions (CBF), Quadratic Programming (OSQP), Pure Pursuit, B-Spline Path Planning, Gazebo Harmonic/Classic.
Perception & SLAM:	LiDAR SLAM (GMapping, SLAM Toolbox), ORB-SLAM2/3, Point Cloud Library (PCL), Foundation Depth Models (Depth Anything V2, UniDepth), OpenCV, YOLOv8, Sensor Fusion (EKF, Particle Filters).
Languages & Frameworks:	C, C++, Python, PyTorch, TensorFlow, MATLAB, Linux (Ubuntu), Git, Docker, L ^A T _E X, Bash scripting, RViz2.
Hardware & Embedded:	NVIDIA Jetson Orin Nano, Crazyflie 2.1 nano-UAV, Velodyne VLP-16 LiDAR, Livox 4D LiDAR, Intel RealSense D435i, UWB Loco Positioning, IMUs, STM32, Arduino, ESP32, Altium Designer, EasyEDA.

ACADEMIC REFERENCES

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Assistant Professor

Dept. of Aerospace Engineering

IIT Madras, India

Thesis Co-Advisor & Project Guide

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Dr. Ram Prasad Padhy

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IIT Bhubaneswar, India

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